

2080

Bachelor of Science in Computer Science and Information Technology

Course Title: **Advanced Database**

Code No: **CSC461** Semester: **8th Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Explain different features of enhanced ER (EER) model. Compare specialization with generalization.
2. Compare object-oriented database with object relational database. What are different object database features that have been included in SQL3?
3. Why distributed database is important? What is transparency? Explain different types of transparencies in distributed databases.

Section B

Attempt any Eight questions[8×5=40]

4. Why hashing is important to store data in databases? What is primary file organization?
5. Explain aggregation with suitable example.
6. What is ODMG object model? What is object definition language (ODL)?
7. Explain different steps in query processing.
8. Why query optimization is essential in databases? What is heuristic optimization?
9. Define fragmentation. Explain horizontal fragmentation with example.
10. What are the characteristics of NOSQL systems? Explain.
11. What is the concept of active database? What are triggers?
12. Write short notes on:
a. BigDatab. Information retrieval